Cloud Based File Security System with Hybrid Encryption

PPD-II First Review | PID: 23CSEP03

PID: 23CSEP03

Ashish Kumar (2301341013)

Ashish Singh (2301341014)

Avee Ranjan (2301341016)

Ayshi Shannidhya Panda (2301341017)

Under the guidance of
Mrs. Snehanjali Pradhan

Silicon Institute of Technology

August 2026

Silicon
...beyond teaching

1 Problem Identification
2 Objectives
3 Literature Review
4 Research Gap
5 Methodology
6 Work Plan
7 References

Silicon
...beyond teaching

Problem Identification

- Cloud storage faces data breaches and unauthorized access
- Data is vulnerable to interception during transmission and compromise at rest
- Single encryption is insufficient — AES has key distribution issues; RSA is too slow for large files

Silicon ...beyond teaching

Objectives

Silicon
...beyond teaching

- Develop a **hybrid encryption system** combining AES and RSA
- Implement **user authentication** with secure password hashing
- Enable **encrypted file upload/download** with client-side encryption
- Design **secure key management** for key generation, storage, and distribution
- Implement **controlled file sharing** via public-key cryptography

Silicon
...beyond teaching

Literature Review

- **Subashini & Kavitha (2011)**: Surveyed cloud security issues; identified data confidentiality as a major challenge
- **Kamara & Lauter (2010)**: Proposed client-side encryption with secure key wrapping for cloud storage
- **Hashizume et al. (2013)**: Analyzed cloud vulnerabilities; recommended layered security approaches
- **Singh et al. (2012)**: Highlighted the need for combining symmetric and asymmetric encryption
- **Kumar et al. (2019)**: Concluded client-side hybrid encryption offers strongest cloud protection
- **NIST SP 800-57 (2020)**: Recommends AES-256 and RSA-2048 for secure systems

Silicon ...beyond teaching

Research Gap

- Most systems rely on **server-side encryption** — provider controls keys
- Hybrid encryption studies are mostly **theoretical**, lacking working implementations
- Few systems integrate **encryption, authentication, key management, and sharing** together
- Existing tools require **manual key management**, leading to human error
- File sharing depends on **password sharing** rather than cryptographic key exchange

Silicon
...beyond teaching

Methodology

How Do We Solve This Problem?

- **Hybrid encryption**: AES for fast data encryption + RSA for secure key exchange
- **Client-side encryption**: files encrypted before upload
- **User authentication** with secure password hashing
- **Automated key management**: generation, storage, and distribution
- **Controlled file sharing** via RSA key re-encryption

Silicon ...beyond teaching

System Architecture

System Components	
Client Application	Login, file upload, download, sharing
Authentication Module	Registration, login, session management
Encryption Engine	AES (data encryption) + RSA (key wrapping)
Key Management	Key generation, storage, distribution
Cloud Backend	REST API, database, encrypted file storage

Silicon
...beyond teaching

Hybrid Encryption Workflow

Encryption (Upload)

1. Generate random AES session key
2. Encrypt file data using AES
3. Encrypt AES key using user's RSA public key
4. Upload encrypted file + encrypted key to server

Decryption (Download)

1. Download encrypted file + encrypted AES key
2. Decrypt AES key using user's RSA private key
3. Decrypt file data using AES key $\rightarrow$ **Original File**

Silicon ...beyond teaching

Controlled File Sharing

Sharing Mechanism

1. Owner decrypts AES key with their RSA private key
2. Re-encrypts AES key with **recipient's RSA public key**
3. Recipient decrypts AES key with their RSA private key
4. Recipient decrypts file using AES key

Security Advantage

No password or private key is ever shared. Only RSA-encrypted session keys are exchanged.

Silicon
...beyond teaching

Work Plan

Silicon
...beyond teaching

- **Phase 1**: Literature survey, requirement analysis, system design
- **Phase 2**: Hybrid encryption engine (AES + RSA)
- **Phase 3**: User authentication and key management
- **Phase 4**: Cloud backend — REST API, database, file storage
- **Phase 5**: Encrypted file upload and download
- **Phase 6**: Controlled file sharing
- **Phase 7**: Testing, validation, and documentation

Silicon ...beyond teaching

References

Silicon
...beyond teaching

References

[1] S. Subashini and V. Kavitha.
A survey on security issues in service delivery models of cloud computing.
Journal of Network and Computer Applications, 34(1):1–11, 2011.

[2] Ronald L. Rivest, Adi Shamir, and Leonard Adleman.
A method for obtaining digital signatures and public-key cryptosystems.
Communications of the ACM, 21(2):120–126, 1978.

[3] Morris J. Dworkin.
Recommendation for block cipher modes of operation: Galois/Counter mode (GCM) and GMAC.
In *NIST Special Publication 800-38D*. National Institute of Standards and Technology, 2007.

[4] Alex Biryukov, Daniel Dinu, and Dmitry Khovratovich.
Argon2: New generation of memory-hard functions for password hashing and other applications.
In *IEEE European Symposium on Security and Privacy (EuroS&P)*, pages 292–302. IEEE, 2016.

[5] Elaine Barker.
Recommendation for key management: Part 1 – general.
In *NIST Special Publication 800-57 Part 1, Revision 5*. National Institute of Standards and Technology, 2020.

Silicon ...beyond teaching

Thank You

Silicon
...beyond teaching